Exercise 2

Exercise 2 — JUGA S.A., opening balance sheet

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> *The JUGA S.A. firm is started with a shareholders' contribution of 60,000 €. Of those, 30,000 € are paid in and deposited in a bank account, and the remaining 30,000 € correspond to the transfer of a patent. With the cash of the bank account, raw materials are purchased for 10,500 € and machinery for 15,000 €. Detail the balance sheet at the start of operations.*

Phase 1 — Setup: data and hypotheses

- The capital contributed is **60,000 €**: 30,000 € in money and 30,000 € in kind (a patent).
- The purchases are paid **with the bank account** (no debt is created).
- It is the **start** of operations: nothing has been produced or sold yet, so there are no revenues or expenses and no income statement.

Phase 2 — Frames: effect of each operation on the balance sheet

Operation	Assets	Liabilities	Equity
Money contributed	Bank +30,000		Capital +30,000
Patent contributed	Patent +30,000		Capital +30,000
Raw materials bought	Raw materials +10,500; bank −10,500		
Machinery bought	Machinery +15,000; bank −15,000		

Each row keeps Assets = Liabilities + Equity [Slide 12]: the contributions increase both sides; the purchases only change the composition of the assets.

Phase 3 — Derivation

Step 1. Bank balance

$$\text{Bank} = 30{,}000 - 10{,}500 - 15{,}000 = 4{,}500\ €$$

Step 2. Classification

- **Patent:** non-current asset [Slide 15].
- **Machinery:** non-current asset (equipment) [Slide 15].
- **Raw materials:** inventory, a current asset [Slide 14]; not consumed yet, so not an expense.
- **Bank:** cash, a current asset [Slide 14].
- **Share capital:** contributed capital [Slide 17].

Step 3. Balance sheet at the start of operations

Assets	**€**	**Liabilities and equity**	**€**
Patent	30,000	Share capital	60,000
Machinery	15,000	**Equity**	**60,000**
Non-current assets	**45,000**	**Liabilities**	**0**
Raw materials	10,500		
Bank	4,500		
Current assets	**15,000**		
Total assets	**60,000**	**Total liabilities and equity**	**60,000**

Phase 4 — Interpretation and checks

- **Balance:** $45{,}000 + 15{,}000 = 60{,}000 = 60{,}000 + 0$.
- **Typical mistake:** keeping the bank at 30,000 € while also adding the machinery and the raw materials gives assets of 85,000 € against 60,000 € of equity. Spending cash on an asset never changes the total.
- **Financial structure** [Slide 36]: the company is financed 100% with equity ($L/E = 0$).

Check against the official key

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The balance sheet coincides with the key in every line: bank 4,500; patent 30,000; machinery 15,000; raw materials 10,500; share capital 60,000; total 60,000. The bank movements (+30,000, −15,000, −10,500) are the same.